RLC Second-Order Analysis

ENGS152 Circuits Lab

6 March 2026

Step 1: RLC Calculations

We start by finding the personal resonant frequency f_0

$$f_0 = (surname\ length\ 1 \times name\ length\ 1)^2 + (surname\ length\ 2 \times name\ length\ 2)^2$$

$$f_0 = (5 \times 10)^2 + (6 \times 11)^2 = 6856\ Hz$$

Allowed 10% range: 6856 $\pm$ 0.1(6856) = 6856 $\pm$ 685.6 $\Rightarrow [\mathbf{6170.4}, \mathbf{7541.6}]\ \boldsymbol{Hz}$

Let **C = 0.22uF** and **L = 2.2mH**

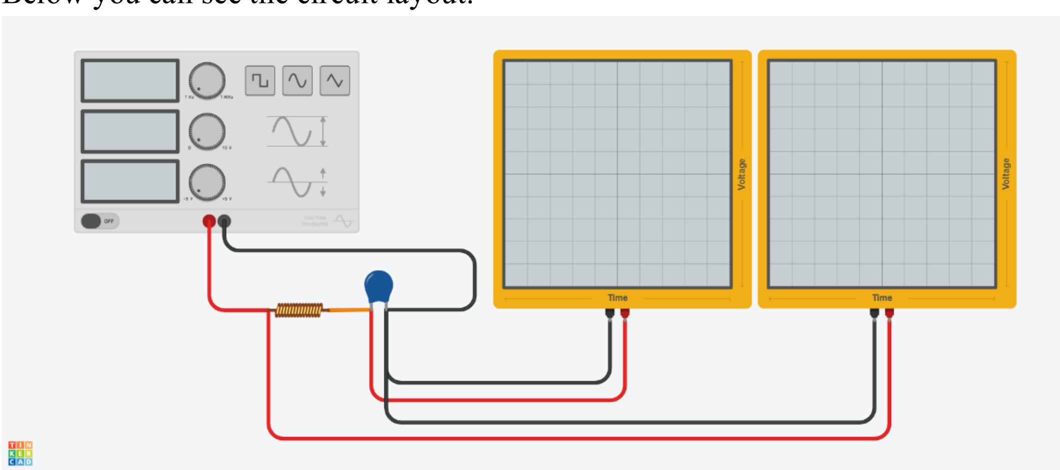
$f_0 = \frac{1}{2\pi\sqrt{LC}} \approx \mathbf{7234.3}\ \boldsymbol{HZ}$**,** the value is within 10% uncertainty.

Quality factor is **10±0.5**:

Real-life output values will be much lower than theoretical calculation; hence there is no need to add a resistor.

Step 2: Circuit Design

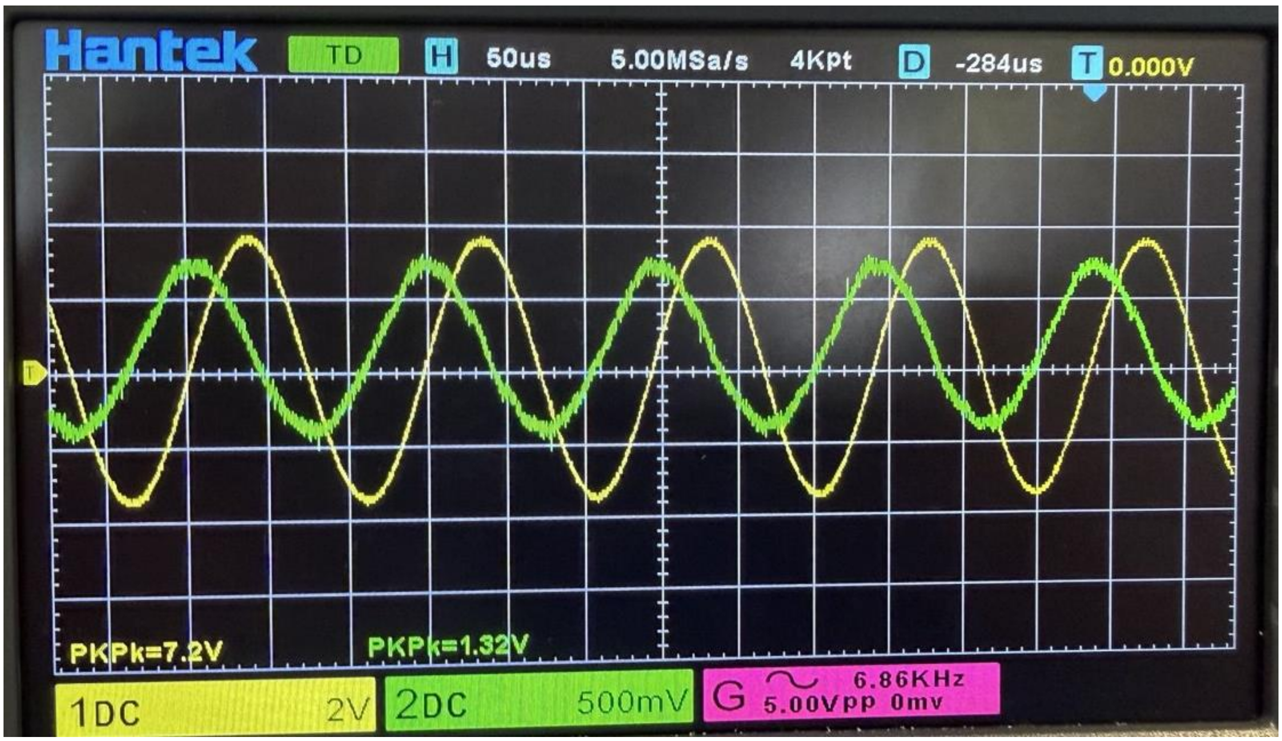
Below you can see the circuit layout:



First, we are asked to generate a **sine** wave with an amplitude of **5V** and use the calculated resonant frequency of $f_0 = \frac{1}{2\pi\sqrt{LC}} \approx \mathbf{7234.3}\ \boldsymbol{HZ}.$

!!!Please note that during the lab we mistakenly used an initial frequency value of 6856 Hz instead of the correct value of 7234.32 Hz, which corresponds to the selected L and C values. This was our mistake, and we realized it as soon as we started preparing the report. All further steps have been taken for the frequency of 6856 Hz. So, it might be misleading. Thank you for your understanding!!!

After supplying the generated signal to the LC circuit and applying the calculated resosnant frequency, two probes were used to visualize the input and output signals. Below you can see the visualization.



Green probe is the input, and yellow probe the voltage across the capacitor, aka the output.

The magnitudes are 1.32 V for the input and 7.2 V for the output. It is an LC Oscillator circuit, so this behavior was expected on the resonant frequency setting. The signal is getting **amplified**.

Next, we are asked to fine tune the frequency such that we find a frequency at which the output is the highest. This is going to be the real-life resonant frequency. After fine tuning it we observed the highest voltage value to be **7.28 V** at frequency of **6850 Hz**. You can see it below.

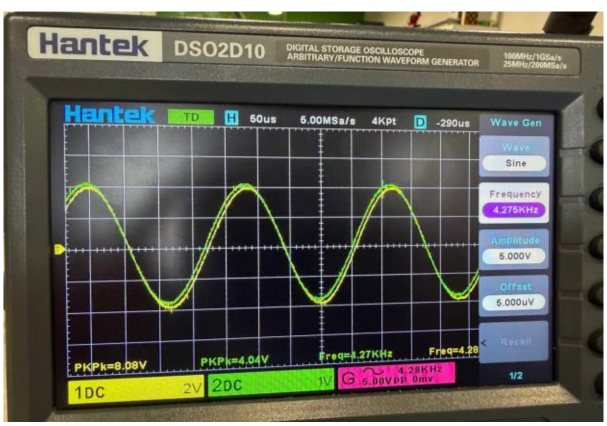
We are asked to calculate what LC should be to get the observed real-life resonant frequency.

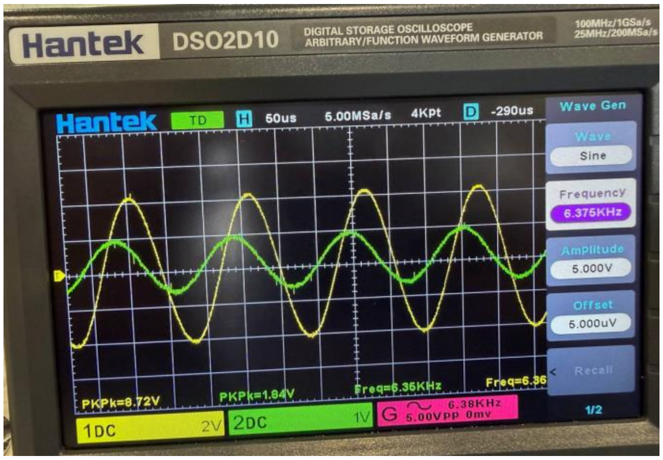
$f_0 = \frac{1}{2\pi\sqrt{LC}} \Rightarrow LC = \frac{1}{4\pi^2 f_0^2} = \mathbf{5.39 * 10^{-10}}\ s^2$

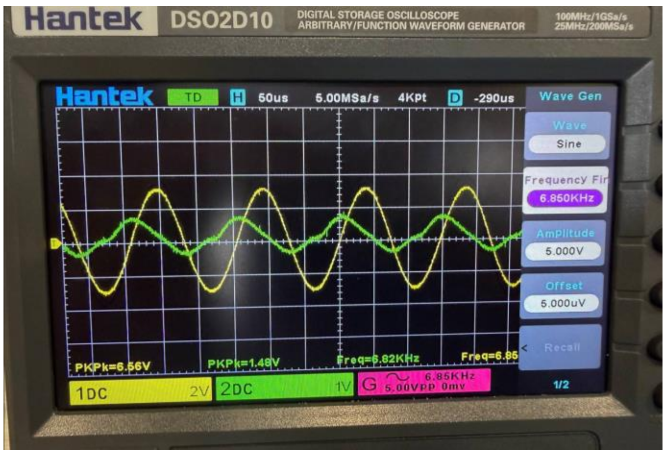
After assuming the capacitor's performance coincides with the theory. Inductor's value must be $L = \frac{5.39\times10^{-10}}{0.22\times10^{-6}} = \boldsymbol{2.45mH}$, it has a percent error of 11.36% from the chosen value of **2.2mH**.

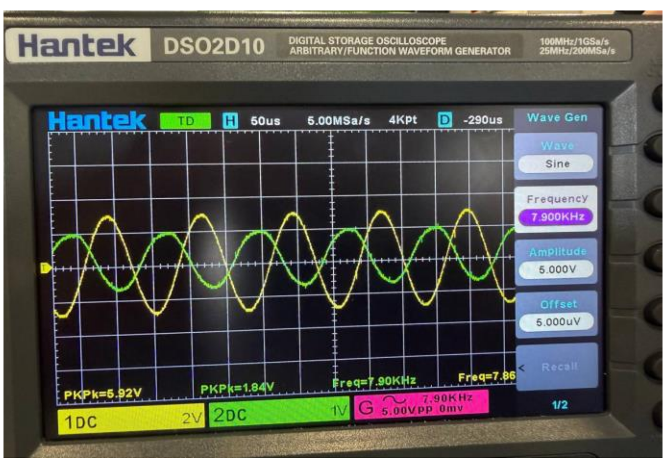
After assuming the inductor's performance coincides with the theory. Capacitors' value must be $C = \frac{5.39\times10^{-10}}{2.2\times10^{-3}} = \boldsymbol{0.24uF}$, it has a percent error of 9.09% from the chosen value of **0.22uF**.

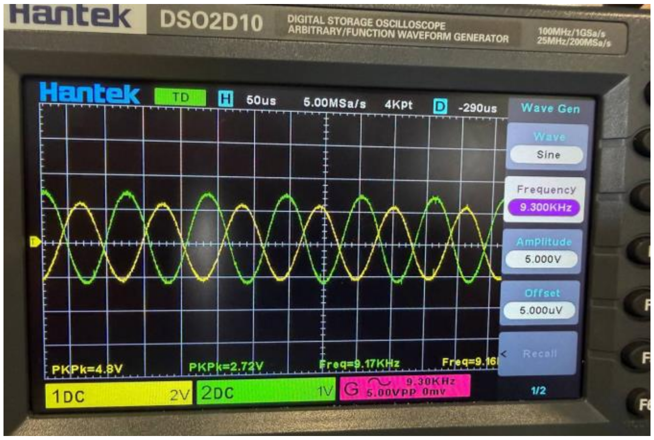
Step 3: Phase shift at different frequencies




Hantek DSO2D10
DIGITAL STORAGE OSCILLOSCOPE ARBITRARY/FUNCTION WAVEFORM GENERATOR
TD 50us 5.00MSa/s 4KPt -290us
Wave Gen
Wave Sine
Frequency 6.375KHz
Amplitude 5.000V
Offset 5.000uV
Recall
PKPk=8.72V
PKPk=1.84V
Freq=6.35KHz
Freq=6.36
6.38KHz 5.00VPP 0mV
1DC 2V
2DC 1V
1/2


Hantek DSO2D10
DIGITAL STORAGE OSCILLOSCOPE ARBITRARY/FUNCTION WAVEFORM GENERATOR
TD 50us 5.00MSa/s 4KPt -290us
Wave Gen
Wave Sine
Frequency
6.850KHz
Amplitude 5.000V
Offset 5.000uV
Recall
PKPk=6.56V
PKPk=1.48V
Freq=6.82KHz
Freq=6.85
6.85KHz 5.00VPP 0mV
1DC 2V
2DC 1V
1/2


Hantek DSO2D10
DIGITAL STORAGE OSCILLOSCOPE ARBITRARY/FUNCTION WAVEFORM GENERATOR
TD 50us 5.00MSa/s 4KPt -290us
Wave Gen
Wave Sine
Frequency 7.900KHz
Amplitude 5.000V
Offset 5.000uV
Recall
PKPk=5.92V
PKPk=1.84V
Freq=7.90KHz
Freq=7.86
7.90KHz 5.00VPP 0mV
1DC 2V
2DC 1V
1/2

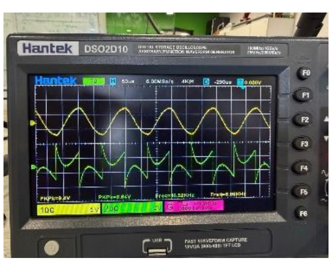


These are the pictures of the phase shift at different frequencies. We can see that when the frequencies are **below resonance** $(f < f_0)$, the phase shift is small, and the output is nearly in phase with the input. At low frequencies, the capacitor's reactance is dominant, meaning the circuit behaves mostly capacitively, and the voltage across the capacitor stays aligned with the source. When the frequency is **close to the resonance,** we can see that the output voltage lags the input approximately by 90^0. Here the resistance of the capacitor and the inductor kind of cancel each other out which results in a phase shift like we got. When the frequency is **above the resonance** the phase shift increases toward 180 degrees, the output kind of becomes "inverted" compared to the input. This is the result of the high inductor resistance compared to the capacitor's.

Step 4: Other types of waves

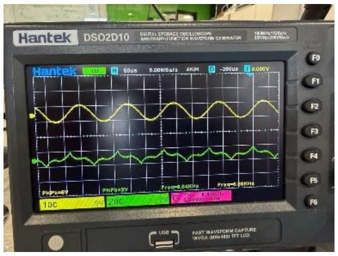
In the end we are asked to experiment with other types of waves and describe the output change. **Square, ramp** and **exponential** waves have been generated to see the output changes.

Square Input



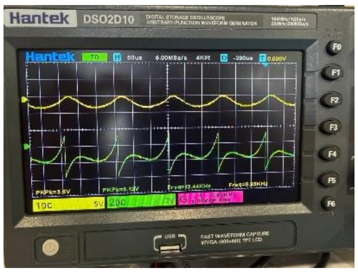
The output voltage becomes almost sinusoidal. $V_c > V_{in}$.

Ramp Input



The output becomes a sine wave. $V_c > V_{in}$

Exponential Input



The capacitor voltage naturally oscillates. $V_c \approx V_{in}$

!!!As mentioned previously these final comments about the output plot might be incorrect because we mistakenly selected wrong resonant frequency!!!